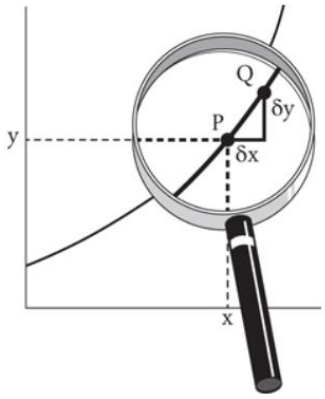
Deep Learning

Quick Start Guide to Calculus

Acknowledgements: Material Taken from 'The Calculus Story' by David Acheson and Pauls Notes online.

Welcome to Deep Learning! This new course is designed to provide insight into the fundamental concepts behind and into the practical applications of Deep Learning frameworks, architectures and applications today.

This is a short revision of some of the calculus that we'll be needing for the course, quick and easy. All of who have done calculus at some point before, this is just to make sure we're all on the same plane moving forward. If you're well versed with all of this, it'll be a very short read! (Do it still)



To begin, what is a derivative? Derivatives allow us to see the behavior of a function (seen as a curve here) at any given point in time. We take two points on the function and find the difference between them, on the x and y axis. We take a small change in x (the independent variable) and check what the resulting change in y (the dependent variable) is over the same very close 2 points.

These small changes are taken to be 'limits' to infinity, meaning that the changes are over an infinitely small range. This allows us to see the specific behavior of a graph at a very specific point.

The whole process of differentiation involves finding the *limit* of $\delta y/\delta x$ as $\delta x \rightarrow 0$, i.e. as Q moves closer and closer to P

Where δy is a small change in y and δx is a small change in x. The resulting $\delta y/\delta x$ gives us the slope or the behavior of a function at a certain point on the independent variable range (x). dy/dx is the notation we use to represent this small change.

$$\frac{dy}{dx} = \lim_{\delta x \to 0} \frac{\delta y}{\delta x}$$

Formally:

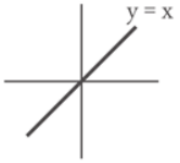
Interpretation of the Derivative

If $y = f(x)$ then,

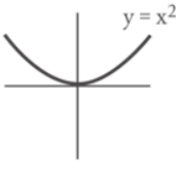
1. $m = f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$ and the equation of the tangent line at $x = a$ is given by $y = f(a) + f'(a)(x - a)$.
2. $f'(a)$ is the instantaneous rate of change of $f(x)$ at $x = a$.
3. If $f(x)$ is the position of an object at time x then $f'(a)$ is the velocity of the object at $x = a$.

Where y is a function f, m is the derivative f' of the function f

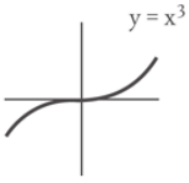
To give a more concrete picture, here are some graphs of functions and their derivatives (behavior)



$\frac{d}{dx}(x) = 1$



$\frac{d}{dx}(x^2) = 2x$



$\frac{d}{dx}(x^3) = 3x^2$

Where the derivatives give us the trend of our function at any given point on the x range.

There are designated rules to be able to calculate the derivatives of different functions that contain different mathematical functions and constructs, here are some.

Basic Properties and Formulas

If $f(x)$ and $g(x)$ are differentiable functions (the derivative exists), c and n are any real numbers,

1. $(cf)' = cf'(x)$
2. $(f \pm g)' = f'(x) \pm g'(x)$
3. $(fg)' = f'g + fg'$ – **Product Rule**
4. $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ – **Quotient Rule**
5. $\frac{d}{dx}(c) = 0$
6. $\frac{d}{dx}(x^n) = nx^{n-1}$ – **Power Rule**
7. $\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)$ This is the **Chain Rule**

Ex.1 – Basic power rule

$$f(x) = 15x^{100} - 3x^{12} + 5x - 46$$

$$\begin{aligned} f'(x) &= 15(100)x^{99} - 3(12)x^{11} + 5(1)x^0 - 0 \\ &= 1500x^{99} - 36x^{11} + 5 \end{aligned}$$

Simply follow this **power rule** when using the product and quotient rule.

Ex.2 Basic chain rule

For the chain rule, we deal with functions composed together, no rule exists to deal with:

$$R(z) = \sqrt{5z - 8}$$

So, we break this down into two sub expressions and use them to compose our R function.

$$f(z) = \sqrt{z} \qquad g(z) = 5z - 8$$
$$R(z) = (f \circ g)(z) = f(g(z)) = \sqrt{5z - 8}$$

We can now go ahead and come up with the required sub derivatives of the interacting parts (left) and apply the chain rule to get our required derivative (right)

$$f(z) = \sqrt{z} \qquad g(z) = 5z - 8$$
$$f'(z) = \frac{1}{2\sqrt{z}} \qquad g'(z) = 5$$

$$\begin{aligned} R'(z) &= f'(g(z))\ g'(z) \\ &= f'(5z - 8)\ g'(z) \\ &= \frac{1}{2}(5z - 8)^{-\frac{1}{2}}\ (5) \\ &= \frac{1}{2\sqrt{5z - 8}}\ (5) \\ &= \frac{5}{2\sqrt{5z - 8}} \end{aligned}$$

Ex3. Reciprocal Derivative

You are able to get to this point using the product rule, but the following is simply a shorthand for it.

If g is differentiable at x, and $g(x) \neq 0$,the $\frac{1}{g(x)}$ is differentiable at x and

$$\frac{d}{dx}[\frac{1}{g(x)}] = -\frac{\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

Basic Derivation Rules

These basic rules of derivations apply over all the different possible functions that can include trigonometric and logarithmic functions as well.

Following are a further set of rules for computing the differentials of some specific mathematical functions:

Common Derivatives

$\frac{d}{dx}(x) = 1$	$\frac{d}{dx}(\csc x) = -\csc x \cot x$	$\frac{d}{dx}(a^x) = a^x \ln(a)$
$\frac{d}{dx}(\sin x) = \cos x$	$\frac{d}{dx}(\cot x) = -\csc^2 x$	$\frac{d}{dx}(\mathbf{e}^x) = \mathbf{e}^x$
$\frac{d}{dx}(\cos x) = -\sin x$	$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}(\ln(x)) = \frac{1}{x}, \ x > 0$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}(\ln\lvert x\rvert) = \frac{1}{x}, \ x \neq 0$
$\frac{d}{dx}(\sec x) = \sec x \tan x$	$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$	$\frac{d}{dx}(\log_a(x)) = \frac{1}{x \ln a}, \ x > 0$

Ex.1 Trigonometric

$$\boxed{y = 5\sin(x)\cos(x) + 4\csc(x)}$$

Use product rule on the first term

$$\begin{aligned} y' &= 5\cos(x)\cos(x) + 5\sin(x)(-\sin(x)) - 4\csc(x)\cot(x) \\ &= 5\cos^2(x) - 5\sin^2(x) - 4\csc(x)\cot(x) \end{aligned}$$

Ex.2 Trigonometric

$$\boxed{P(t) = \frac{\sin(t)}{3 - 2\cos(t)}}$$

$$\begin{aligned} P'(t) &= \frac{\cos(t)(3 - 2\cos(t)) - \sin(t)(2\sin(t))}{(3 - 2\cos(t))^2} \\ &= \frac{3\cos(t) - 2\cos^2(t) - 2\sin^2(t)}{(3 - 2\cos(t))^2} \end{aligned}$$

Remember that $\cos^2(\theta) + \sin^2(\theta) = 1$.

$$P'(t) = \frac{3\cos(t) - 2\left(\cos^2(t) + \sin^2(t)\right)}{(3 - 2\cos(t))^2}$$
$$= \frac{3\cos(t) - 2}{(3 - 2\cos(t))^2}$$

Ex 3. Logarithmic

$$\boxed{f(x) = 3\mathbf{e}^x + 10x^3 \ln x}$$

Use the product rule on the second term

$$f'(x) = 3\mathbf{e}^x + 30x^2 \ln x + 10x^3\left(\frac{1}{x}\right)$$
$$= 3\mathbf{e}^x + 30x^2 \ln x + 10x^2$$

Ex 4. Logarithmic

$$\boxed{y = \frac{5\mathbf{e}^x}{3\mathbf{e}^x + 1}}$$

Use the quotient rule here

$$y' = \frac{5\mathbf{e}^x\left(3\mathbf{e}^x + 1\right) - \left(5\mathbf{e}^x\right)\left(3\mathbf{e}^x\right)}{\left(3\mathbf{e}^x + 1\right)^2}$$
$$= \frac{15\mathbf{e}^{2x} + 5\mathbf{e}^x - 15\mathbf{e}^{2x}}{\left(3\mathbf{e}^x + 1\right)^2}$$
$$= \frac{5\mathbf{e}^x}{\left(3\mathbf{e}^x + 1\right)^2}$$

And here are several simplifications for applying the chain rule to different situations

Chain Rule Variants

The chain rule applied to some specific functions.

1. $\frac{d}{dx}\left(\left[f(x)\right]^n\right) = n\left[f(x)\right]^{n-1} f'(x)$
2. $\frac{d}{dx}\left(\mathbf{e}^{f(x)}\right) = f'(x)\mathbf{e}^{f(x)}$
3. $\frac{d}{dx}\left(\ln\left[f(x)\right]\right) = \frac{f'(x)}{f(x)}$
4. $\frac{d}{dx}\left(\sin\left[f(x)\right]\right) = f'(x)\cos\left[f(x)\right]$
5. $\frac{d}{dx}\left(\cos\left[f(x)\right]\right) = -f'(x)\sin\left[f(x)\right]$
6. $\frac{d}{dx}\left(\tan\left[f(x)\right]\right) = f'(x)\sec^2\left[f(x)\right]$
7. $\frac{d}{dx}\left(\sec\left[f(x)\right]\right) = f'(x)\sec\left[f(x)\right]\tan\left[f(x)\right]$
8. $\frac{d}{dx}\left(\tan^{-1}\left[f(x)\right]\right) = \frac{f'(x)}{1+\left[f(x)\right]^2}$

Here are some examples of combining the chain rule with logarithmic functions

Ex. 1 Logarithmic

$$\boxed{g(x) = \ln\left(x^{-4} + x^4\right)}$$

$$g'(x) = \frac{1}{x^{-4} + x^4}\left(-4x^{-5} + 4x^3\right) = \frac{-4x^{-5} + 4x^3}{x^{-4} + x^4}$$

Ex. 2 Logarithmic

$$\boxed{f(t) = \left(2t^3 + \cos(t)\right)^{50}}$$

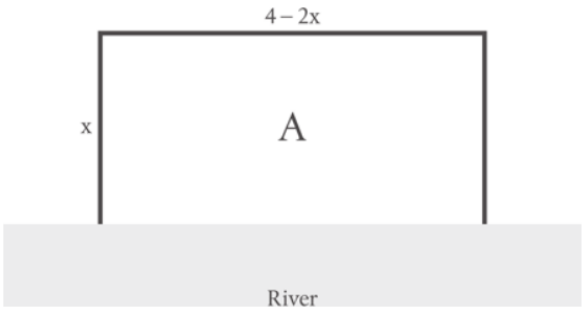
$$\begin{aligned} f'(t) &= 50\left(2t^3 + \cos(t)\right)^{49}\left(6t^2 - \sin(t)\right) \\ &= 50\left(6t^2 - \sin(t)\right)\left(2t^3 + \cos(t)\right)^{49} \end{aligned}$$

Minima, Maxima, Optimization

A Major use of calculus is in **optimization problems** where we have to find an optimal point given a particular function space. We might be looking for a min point or a max point depending on the nature of the problem.

Machine Learning is basically a very large and slightly more complex optimization problem where you try to best model a real-world problem to a mathematical function by finding optimal function parameters.

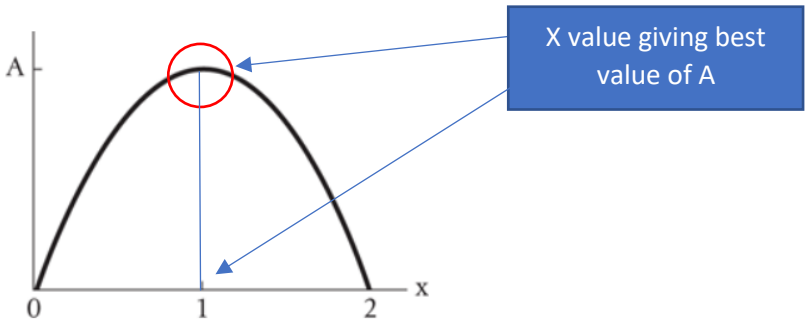
Let's take an example. Pretend you're a farmer, and you want to create a rectangular field next to a river using a fence. You have exactly 4km of fencing for the three non-river sides. How should the dimensions of the fencing be set in order to ensure the largest possible fenced area?



Formally we can define this problem as:

Area = (x)(4 – 2x) where x is the width.

Now that we have a function to define our problem formally, we want to be able to use calculus to solve this '**Maximization**' problem. If we were to sketch out a graph of the problem where A depends on the x, it'd look something like this:



Since we have a very simple problem at hand, we're able to draw this graph out very easily, but if the problem looked more like

$$A = x^4 + \frac{1}{2}\sqrt{x}^3 + 37\ln x$$

We'd be hard pressed to graph this, let alone scan the full range of values for the visual max point. This is where differentiation comes into its own and takes care of this for us. We can see that a max/min point has an instantaneous derivative or function behavioral trend of 0.

Meaning, if we differentiate the area function, we have a way to map its behavior at different x points. Set this to 0 and it will provide the 'x' points at which the function is stationary, or at an inflection point.

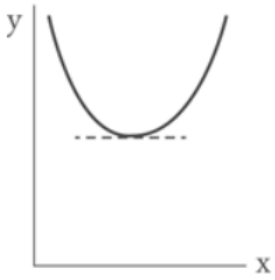
This is then a candidate for an optimal (min or max) point.

$$\frac{dA}{dx} = 4 - 4x. \quad \Longleftrightarrow \quad \frac{dA}{dx} = 0,$$

The result is the best value of x = 1, the same as we visually saw. This leads to a max A of 2. Optimal.

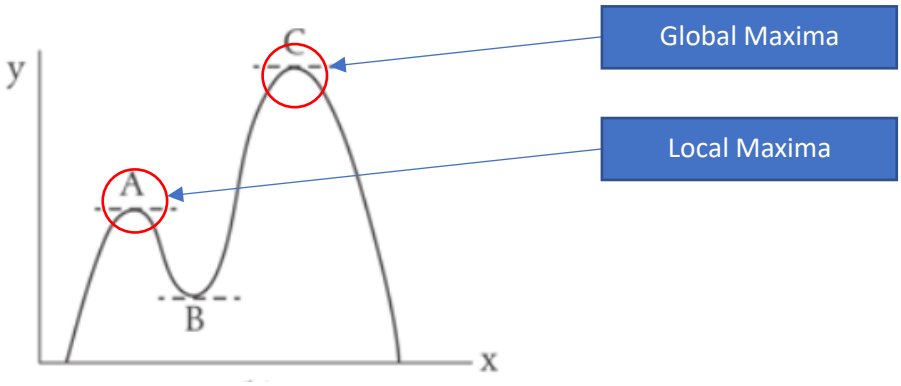
The Optimization Caveat

Just as we figured out the optimal point in a maximization problem, we can find out optimal points to minimization problems as well. One that might be defined in 2d space as follows:



Again, we can see that the point of interest is where the graph is flat, with no instantaneous trend / derivative 0.

The caveat with optimization problems is that, given any real-world problem, the defining functions are rarely this simple. Following is a slightly more complex example, with 3 'stationary points' all of which have a gradient of 0, (function does not have increasing or decreasing trend). This is where hunting for optimal points suddenly has to incorporate dealing with 'local' and 'global' minimas and maximas.



Once we define our deep learning problems as a hunt for global minimas in higher dimensional spaces, they have to hunt around function spaces (much like the graph above), trying to avoid getting stuck in local minimas, which would result in a sub optimal solution. This will be discussed at length in class.

Practice Problems

You can find a plethora of practice questions online but here are some to get you started. Doing these is highly encouraged in order to get comfortable with differentiation different kinds of functions, they tend to show up everywhere in the field of deep learning. The more comfortable you are, the easier it'll be to fly through new material.

Differentiate the following:

$$y = 6 + 4\sqrt{x}\,\csc(x)$$

$$f(w) = \tan(w)\sec(w)$$

$$R(t) = \frac{1}{2\sin(t) - 4\cos(t)}$$

$$g(t) = 4\log_3(t) - \ln(t)$$

$$y = z^5 - \mathbf{e}^z \ln(z)$$

$$f(t) = \frac{1 + 5t}{\ln(t)}$$

$$f(t) = 5 + \mathbf{e}^{4t + t^7}$$

$$F(y) = \ln\left(1 - 5y^2 + y^3\right)$$

- Hint -> Last 2 require chain rule, the previous are a mix of the more basic rules all provided before.
- We'll give you the solutions later, just make sure you try all of them, even if you don't get them completely